

Forecasting Monthly County Voter Registration Totals

Carl E. Klarner, klarnerpolitics.org and Thessalia Merivaki, Mississippi State University

Abstract

A model is developed to forecast monthly county level voter registration totals by party for Florida, Iowa, Maryland, Oregon and Wyoming. A prediction model of change in voter registration can help election administrators prepare for periods of high workload. Another application is to aid the adjustment of voter registration figures with an eye towards removing deadwood registrants. The aim of doing so is to boost voter registration data's efficacy in predicting phenomena such as postcensal population growth, election outcomes and voter turnout. Descriptive statistics on the concentration of drops and negative net changes in total registrants are presented to discuss how much information such data can yield to help estimate deadwood in conjunction with prediction models utilizing demographic and other factors.

Keywords: voter registration; list maintenance; deadwood; NVRA

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Introduction

To address questions of data management integrity, equitable access to the voting process, potentially discriminatory practices in election administration, but also electoral dynamics in voter registration, scholars have relied on two sets of data: the Election Administration and Voting Survey (EAVS) county-level data, made available bi-annually, and/or individual-level data drawn from states' voter registration records. These analyses have uncovered important pieces of the voter list maintenance puzzle, particularly the quality of voter registration lists, and the quantity of removed voter registrants across and within the states, as well as across time. Furthermore, aggregate voter registration statistics have the potential to direct mobilization efforts, assess reforms aimed at boosting participation in elections and aiding post-decennial population estimates by the Census.

Yet, they have some limitations, such as the reliability and timeframe of county-level data as reported in the EAVS (Merivaki 2020; Stewart III 2019; Merivaki and Smith 2015), and the availability of individual voter records beyond a select number of states due to the high costs of obtaining them (McDonald, Licari and Merivaki 2015). In lieu of bi-annual data, analysis of monthly patterns in county-level voter registration totals can provide more accurate insight into the administration of voter registration records, for example, by evaluating how “even” list maintenance is from county to county. Voter registration statistics by party can also be used for election forecasting.

Voter registration data are particularly useful because of shortcomings with surveys of the general population about their voting behavior and registration status. Putting aside problems with sample and response bias, which are well known (see Ansolabehere and Hersh 2012; McDonald 2007), they offer answers to questions about the administration of elections that public opinion survey data cannot answer. That said, the limitations of using total registrants are well understood (Merivaki 2020; Stewart III 2019). But when data collection at the preferred granular level is prohibitively expensive, it is worth examining how much can be squeezed out of readily available aggregate data.

A potential challenge when learning from voter registration statistics is the presence of deadwood registrants--people who are no longer eligible to vote in the area because of death, having moved, committed a disenfranchising crime, etc. It is the aim of this study to adjust voter registration data with an eye towards removing deadwood registrants to increase its efficacy as a predictive tool for researchers and election administrators. Such adjustments may very well not give us the level of certainty one would need to make assertions--normative or otherwise--about the absolute number of deadwood registrants in a particular place and time (Stewart III 2019). But the efficacy of any adjustments for making forecasts can be evaluated empirically, and such adjustments can also give us further insight into the efficacy of the list maintenance process across and within the states.

When voters are removed from the rolls, it technically means that at the time of removal they were no longer eligible to vote. (Note that this statement does not imply that it is not a major problem from a normative standpoint that a small fraction of the removed are eligible.) There is often a delay between when voters became ineligible and when they were removed, as by federal law (NVRA) the removal process is long and requires a series of steps which span over at least two federal elections. The distribution of the length of this delay presumably varies greatly over time and place due to federal and state election laws, election administration procedures, as well as electoral factors. A precise assessment therefore is difficult, although doing so would be extremely informative.

Aggregate analyses of removed voters strongly suggest that there is clustering in time regarding when people are dropped from the rolls (Feder and Miller 2020; Morris 2020). It is likely that the delay associated with being removed from the voter records is greater on average than when clustering is not present. Election administrators, especially in small jurisdictions and with fewer resources, presumably concentrate their list maintenance activities to take advantage of economies of scale. Often, they are constrained by state law on when to conduct voter list maintenance (Merivaki 2020). They also have to avoid the prohibition on removing people from the rolls for non-voting (together with not replying to a confirmation letter) 90 days before a federal election and they may not use fewer than two general elections in a row to do so, as the National Voter Registration Act (NVRA) outlines.

Having aggregate data on both removals and the addition of new registrants by month allows the assessment of this concentration better than if the total number of registrants is observed over a longer period of time. However, information about removals can be gleaned from the net change in total registrants, and what can be gleaned increases as negative change is more concentrated in time. Estimates of drop activity and its concentration can also be made more accurate by utilizing a model predicting total voter registrants. When the activity of new voter registrations, or adds, is concentrated, it especially improves the accuracy of estimates of the number of adds, which can then be subtracted from the total to yield an estimate of drops.

The Determinants of Voter Registration Management

The management of voter registration is mainly understood to revolve around the removal of ineligible voters from voter rolls (Morris 2020; Feder and Miller 2020; Stewart III 2019; Biggers and Smith 2018; Herron and Smith 2018; Pettigrew 2017), and the accuracy of voter registration records (Merivaki 2020; Shino et al. 2020; Kim, Cao and Alvarez 2020; Kim, Schneider and Alvarez 2019; Ansolabehere and Hersh 2010; Ansolabehere, Hersh, Gerber and Doherty 2010). What is broadly defined as voter list maintenance is regulated by the National Voter Registration Act (NVRA) of 1993 and the Help America Vote Act (HAVA) of 2002. The former prohibits states from removing registered voters for non-voting up to 90 days prior to a federal election, and explicitly forbids the removal of voters due to non-voting without sending out a letter to confirm they are no longer at the address listed in their record. HAVA mandated that states adopt statewide, electronic voter registration databases to increase the accuracy of individual voter record maintenance. Both federal laws aimed to increase access to voter registration and improve the efficiency of voter registration management.

The *Shelby v. Holder* SCOTUS decision brought a renewed interest in how voter records are managed, with evidence that states and localities previously covered under the VRA's preclearance requirement engage in more aggressive voter removal rates (Feder and Miller 2020; Morris 2020; Hardy 2020), as well as utilizing "creative" ways to go around the NVRA's legal restrictions for voter removals (Herron and Smith 2018). Research from provisional voting connects voter list maintenance to voter removals, as there are voters who were not able to cast a regular ballot because they were previously removed (Merivaki and Smith 2020; Morris 2020).

A point of contention is whether these removals were unlawful because they violate the NVRA (Herron and Smith 2018), whether they were “wrongful” because voters were required to vote provisionally (Morris 2020), or resulted from mistakes in how active and inactive records are managed by election officials (Merivaki 2020; Kim, Schneider and Alvarez 2019). In the latter case, researchers find that such errors may be corrected in the provisional ballot verification stage (Merivaki and Smith 2020).

Voter registration management is a complex enterprise, as it often includes adding new voters and updating duplicate registrations on a daily basis. It involves frequent, if not real-time, data sharing between federal, state and local agencies in order to determine whether a voter has died, moved to a different jurisdiction, or moved into a new locality. It involves updating voter records in multiple intervals from active to inactive. There is variation across and within the states in which procedures are used to manage voter records, with limited guidance from the NVRA (Merivaki 2020). Research suggests that due to the complexity in managing new and ineligible voter registrants, voter registration records often appear inflated because of the number of inactive voters who may not have yet been removed, and as a result voter registration totals reported in the bi-annual EAVS are unavoidably exaggerated (Stewart 2019 III).

Because voter registration management is an ongoing process, we lose a lot of the nuances in how Americans register to vote, as well as the challenges for election administrators to efficiently manage voter registration, when looking only at voter registration totals reported within a window of two years. Aggregate patterns of voter registration make it clear that voter registration rates decrease dramatically between a presidential and a midterm election. This has been to some extent attributed to increased voter interest for registration in a presidential election cycle, but there are administrative reasons to explain why voter registration can dramatically drop during the time period between a presidential and a midterm election (Merivaki 2020). Between that time period, for instance, there may be special elections, or primary elections. If these elections are held for federal office, then election administrators are restricted from “cleaning” voter registration records, as mandated by the NVRA.

But, the addition of new voters is ongoing, and so is the process of updating voters who move from one jurisdiction to another (Stewart III 2019), or change their partisan affiliation

(Cao, Kim, and Alvarez 2020). That the rates of new voter registrations increase as presidential elections approach is well known. However, how concentrated new registrations are in the months before elections and how this concentration varies from locale to locale and by party is not. Furthermore, it is not known how much new registration activity increases before primaries relative to the general election and how much the competitiveness of primaries and general elections in a state and year contribute to new registered voters.

Insofar as removing ineligible voters, there seem to be specific windows for election administrators to drop them from the voter rolls, yet at the same time update records from active to inactive. Because non-voting is not sufficient by itself to remove someone from the rolls, states and localities utilize specific “triggering” events to communicate with active registered voters seeking a confirmation that they reside at their residence on record (Merivaki and Smith 2020). In some states, a jury duty summons that is returned as undeliverable can constitute a trigger for election officials to contact a registered voter. One of the most common triggers is the National Change of Address (NCOA), which can be a useful tool in updating active voter records and encourage new voter registrations (Kim 2019).

Voter registration deadlines, which vary by state, can create various windows for voters to register to vote, and thus affect the flow of new voter registrations during an election year. Analysis of monthly-level voter registration data strongly suggest that new voter registrations surge when voter registration deadlines approach (Merivaki 2020; Shino and Smith 2018). If a state is to hold a primary, a runoff, a special election and a general election for federal office during an election year, it follows that these surges will be dependent on the voter registration deadlines prior to these elections. In states where Same Day/Election Day Registration (SDR/EDR) is available, it is often the case that voter registration may follow a more normal distribution, as voters can register to vote up to Election Day. Furthermore, states that offer Online Voter Registration (OVR) may have higher overall new voter registration rates, because the electronic process minimizes the risk of errors and increases the chances of successfully registering to vote (Merivaki 2020; Merivaki and Mann 2019). These dynamics are largely ignored when biennial data are analyzed. Monthly-level data can, therefore, capture these patterns in voter registration flow more accurately. The monthly data also can capture seasonality

in removing ineligible voters, and therefore measure more accurately how voter registration grows and declines over time, relative to the ratio of new and removed voters (Merivaki 2020).

Because in some states a prospective registrant is required to register with a political party, a useful metric to include in the assessment of voter registration management is the change in partisan voter registration over time. Increases in voter registration by party can result from new voter registrations, or from existing registrants who may choose to change their party registration for any reason, or because they want to participate in a competitive primary election. Understanding these changes across and within states are important, as they allow researchers to explain any shifts in a state's/locality's partisan make-up and voting behavior. Because these dynamics are electoral, they are not measured in the EAVS, yet they involve the management of records, and are therefore relevant in the study of voter registration management.

Monthly County Level Voter Registration Data

Voter files provide an opportunity to examine individual level data, which is very powerful, but have a weakness that severely limits analyses that utilize them exclusively: they are often only available for short periods of time. An example of the limitations of voter file data can be found in Cao, Kim and Alverez (2020) who examine Florida voter files from June to December 2016, finding anomalies regarding drops in Citrus, Duval, Holmes, Leon, Sumter, Taylor and Walton counties. But examination of time series of Florida counties makes it clear that counties engage in drops during different months from each other, making looking at a restricted time periods problematic. Some of this is due to differences in the behavior of the population but much is also presumably due to differences in when county administrators work on their lists. An assessment of the quality of voter records using voter files over a complete biennium may be optimal, although the challenges in obtaining multiple snapshots of voter files across multiple years is a time-consuming, and costly endeavor.

To best predict the efficacy of future policy interventions in voter record management and registration modernization, robust time-series of past elections must be utilized to cope with changing background conditions. Reports of voter registration data allow long periods of time to be examined. As reports are more frequent, reported at smaller levels of geography, and can include breakdowns by new/removed, active and inactive voters, and often by party and race,

they allow researchers to address ecological inference problems more appropriately. The current analysis presents the first round of data collected as part of a larger-scale data collection and analysis project.

In this study, we present data for Alabama, Florida, Iowa, Maryland, Oregon and Wyoming. Table 1 reports relevant election laws that pertain to each state. By happenstance, states with lifetime bans on felons voting are overrepresented. These states were selected because the time series of monthly reports on their websites were lengthier than most other states.

Table 1: Six States Data Collected for and Relevant Election Laws

State	Years & Months Covered	Closing Date: Days Before Elect	Online Voter Reg: Month Adopted	Primary Type	Felon Disenfranchisement
AL	2004-11 to 2020-4	14	2/2016	Open	Lifetime
FL	1995-1 to 2016-12	29	5/2015	Closed	Lifetime
IA	2000-1 to 2020-5	EDR since 2007, 10 before that	1/2016	Closed, but can switch on ED	Lifetime
MD	2000-1 to 2020-4	EDR after sample, 5 from 2014 to 2018, 21 before that	1/2012	Closed	Prisoners Only
OR	2001-1 to 2020-4	21	1/2010	Closed	Prisoners Only
WY	2008-2 to 2020-5	0	No OVR	Closed, but can switch on ED	Lifetime

The richness of the data is demonstrated in Appendix A, that reports variables obtained, number of datapoints for each state and other details. The forecasting model does not include

Alabama, as registration by party does not exist there, but Alabama is referred to below as descriptive statistics are presented for that state later in the paper. Data for the six states were obtained for the following time periods without interruption. For all figures we report below, only data appearing within an entirely observed biennium are utilized.

Ambiguities in Data

A systematic overview of monthly voter registration totals from these states uncovers several ambiguities in what the data represent, when they were reported, and how often they are updated before they are reported in the following time period. Inquiries to the states have been made to fully establish what the data in the reports represent, but much still remains to be clarified.

For instance, from 1997 to 2016 Florida reports drops and valid adds in its monthly reports, as does Maryland from 2003 to 2010. One would think that change in monthly totals from one month to the next would be equal to valid adds minus drops in the contemporaneous month, but this is not the case, whatever combination of potentially appropriate comparisons is used. For Florida, there were 457 out of 15,890 county-months where the figures were as expected and for Maryland there were 45 out of 2,668 county-months. It is not clear, therefore, if duplicate registrations are recorded as a separate category, and why they are not reported with the monthly totals.

To better understand how changes in voter registration are affected by the addition and removal of registered voters over time, change in total registrants at the county level was computed from one month to the next, and then regressed on contemporaneous drops and adds, drops and adds from last month, the month before, and the month before that. All of these variables were then deflated by contemporaneous VEP, to remove any correlations merely caused by county size. Note that since every variable is divided by the same amount for one observation, any precise additive relationships will be unchanged. The output from these regressions, done separately for Florida and Maryland, are reported in Table 2.

Table 2: Regressions Predicting Change in Monthly County Registrants with Contemporaneous and Lagged Reported Adds and Drops

	Florida		Maryland	
	b (s.e)	p	b (s.e)	p
Adds	.781 (.026)	.000	1.00 (.022)	.000
Drops	-.310 (.014)	.000	-.872 (.021)	.000
Adds t-1	.179 (.028)	.000	.027 (.024)	.260
Drops t-1	-.069 (.014)	.000	-.020 (.022)	.349
Adds t-2	.065 (.028)	.019	-.019 (.024)	.436
Drops t-2	-.049 (.014)	.001	-.043 (.021)	.046
Adds t-3	-.022 (.026)	.397	-.102 (.022)	.000
Drops t-3	-.035 (.014)	.014	.009 (.021)	.655
Constant	-.002 (.000)	.000	.001 (.000)	.000
S.E.E.	.01079		.00474	
R-Sq.	.10		0.46	
Adj. R-Sq.	.10		0.46	
N	15, 879		4,484	

Reported p-values are two-tailed.

Note: All variables deflated by contemporaneous voter eligible population.

Table 2 indicates that almost none of the variation in change in total registrants from month to month is explained by the model for Florida. Furthermore, the coefficients associated with lagged adds and drops are statistically significant with the correct signs back to two reports prior for adds, and back to three reports prior for drops. This implies that drops and adds are not being added to the totals as they occur. Why this might be the case is very unclear.

Much more of the variation in change is explained in the model for Maryland, with an R-squared of .46. Furthermore, the coefficient on the contemporaneous adds variable is nearly “1” unlike in the model of Florida. Next, the lagged add and change variables are not statistically

significant in this model, with but one exception, implying that changes are being taken into account in the total, just not very reliably.

Florida also reports “confirmation mailings” and “responses” up to 2010, in other words, the number of letters a county sends out in a month to confirm that someone on their list that they have reason to believe is ineligible to vote is not at that address. “Responses” means that the person in question responds and so prevents their name from being removed from the rolls.

Examination of 670 time series plots of each county-biennium in Florida provides insight into what the data represent, but bring up troubling inconsistencies in the data. The number of responses is generally far smaller than the preceding number of confirmation mailings, and there is generally a short delay between spikes in confirmation mailings and responses—the infrequent times that responses occur to an extent large enough to be apparent. Presumably, there would then be a spike in reported removals (aka “drops”) after the spike in confirmation letters, but this is frequently not the case. For example, in Alachua County in 1997, there were over 5,000 confirmation letters sent out in one month, followed by a net drop in total registrants of a similar magnitude the next month, but no corresponding uptick in reported “voter registrations – deleted.” Much the same happened in 2001. Then, in 2003, there was an uptick in confirmation letters, followed the next month by both an uptick in reported drops and a net decrease in total registrants, all three spikes of a similar magnitude. What happened in 2003 is not consistent with what happened in 1997 and 2001.

The examples above occur over and over again in the time series, and lead to the conclusion that not all drops from the rolls are captured by the column in the reports labeled “Voter Registrations – Deleted” (up to December 2009) or “Voter Registrations – Removed” from January 2010 and afterwards. Table 3 provides evidence that confirmation letters are associated with drops that are not captured by the “drop” column. Variables tracking confirmation letters and responses lagged from zero to three periods are added to the model in Table 3 for Florida, while confirmation letters only are so added to Maryland, as responses are not in the Maryland data.

Table 3: Regressions Predicting Change in Monthly County Registrants with Contemporaneous and Lagged Reported Adds and Drops

	Florida		Maryland	
	b (s.e)	p	b (s.e)	p
Adds	.725 (.230)	.000	1.05 (.018)	.000
Drops	-.261 (.016)	.000	-.721 (.019)	.000
Adds t-1	.177 (.031)	.000	.072 (.019)	.000
Drops t-1	-.018 (.016)	.285	-.058 (.019)	.003
Adds t-2	.066 (.031)	.036	-.023 (.019)	.222
Drops t-2	-.029 (.016)	.073	-.014 (.017)	.419
Adds t-3	.008 (.030)	.781	-.052 (.018)	.004
Drops t-3	-.023 (.017)	.154	.037 (.017)	.031
Notices	-.101 (.008)	.000	-.108 (.012)	.000
Responses	-.110 (.046)	.016	-	-
Notices t-1	-.150 (.009)	.000	-.025 (.012)	.038
Responses t-1	-.096 (.045)	.034	-	-
Notices t-2	-.063 (.009)	.000	-.001 (.004)	.823
Responses t-2	.149 (.046)	.001	-	-
Notices t-3	-.042 (.009)	.000	-.006 (.003)	.082
Responses t-3	.006 (.044)	.898	-	-
Constant	-.002 (.000)	.000	.001 (.000)	.000
S.E.E.	.01189		.00366	
R-Sq	.16		.72	
Adj. R-Sq	.16		.72	
N	10,385		2,636	

Reported p-values are two-tailed.

Note: All variables deflated by contemporaneous voter eligible population.

For Florida, all four confirmation notice variables are statistically significant, although the overall explanatory power of the model does not increase that much: the standard error of the estimate barely goes down at all between the two tables. For Maryland, the contemporaneous and lagged confirmation notice variables are statistically significant, and the standard error of the estimate goes down substantially, although there is still a lot of prediction error in the model.

More anomalies were observed that are not discussed in this section. Indicatively, Gilchrist County saw a large number of confirmation mailings in 2007, but recorded no responses, and the net number of registrants did not go down, even though there were no substantial adds at this time. Sarasota County saw a major net decline in total registered voters in late 2007, but there was no corresponding increase in drops or confirmation letters. In Columbia County in 2010, there are large reported drops but no accompanying drop in net registrants.

The conclusion of the above analysis is that it is unclear exactly what are being reported by “deletions” and “removals” (as in Florida) and “removals” (the term used in Maryland). Although there is good evidence that the confirmation mailing totals are tracking removals brought about by confirmation letters, it is still not possible to trace exactly when the confirmation mailings resulted in voters being removed from the rolls and how many of them were, given that the responses occur with varying delays.

Descriptive Statistics: Concentration of Drops

How concentrated are drops when information on drops and valid adds are available? Does this vary from county to county? And if so, which kind of counties have more concentration in their drops? This section focuses on reported totals of “deletions” and “removals,” despite the uncertainty with what these data represent.

Table 4 indicates that dropping is fairly concentrated. The average county in a two-year period did 18.2 percent of their removals in one month, about a third of removals in three months and half of their removals in six months. The figures above do not change appreciably when weighted by VEP. The concentration of adds is, surprisingly, lower than the concentration of drops, but becomes similar when looking at the top six months, perhaps reflecting that new registrations are spread out in the six or so months before the general election.

It would make sense if there was a positive relationship between the concentration of drops in a county from one biennium to the next but this is not the case. Nor is there a relationship between population size and the level of concentration of drops. The answer to the former is because the concentration in drops is so great, that looking at concentration within county-bienniums is too small of a time span.

Table 4: Concentration of Drops in Florida and Maryland

Variable	Obs.	Adds		Drops	
		Mean	Std. Dev.	Mean	Std. Dev.
Herfindahl Index	766	.056	2.1	.092	8.6
Top Month	766	10.9	5.0	18.2	14.3
Top 2 Months	766	19.3	6.8	26.8	15.0
Top 3 Months	766	25.9	7.4	33.4	14.7
Top 4 Months	766	31.8	7.8	39.1	14.2
Top 5 Months	766	37.1	7.9	44.2	13.7
Top 6 Months	766	42.0	8.0	48.9	13.1

We report extreme examples of drops in one-month periods on Table 5 in the Appendix. The most extreme example in a large county is Miami-Dade in July 2009, which reported dropping 123,791 registrants in that month, 18.1 percent of the total drops between 1997 and 2016. The time series of Miami-Dade, however, indicates that there was no corresponding drop in the aggregate number of registrants, and so not much stock should be put into these data.

Forecasting Change in Monthly Voter Registration Totals

The preliminary model utilized to predict total registrants is as follows. It is set up as a standard “within sample” prediction model, without out of sample assessments of how well the model performs. The unit of analysis in the predictor model is the county-report-party category. Reports are generally monthly, but are sometimes more frequent. For ease of presentation, the language below refers to them as “monthly.” Three party categories are utilized: Democrats, Republicans, and “other” registrants (i.e., unaffiliated, or third-party).

The dependent variable is the total number of registrants per 1,000 voting eligible people (VEP). Estimates of prisoners in a county was done with very approximate Census data, and probationers and parolees estimated at the county level from state data by racial group. Citizen voting age population by race and Hispanic status was estimated with Census data. Of the six states examined here, Alabama, Florida, Iowa, and Wyoming disenfranchise former felons for life, which was not taken into account.

An important forecasting principle is to engage in “bottom-up” forecasting when possible (Armstrong 2001). In bottom-up forecasting, forecasts are made for units at as small of a level as possible and are then aggregated up for a larger forecast. For example, instead of directly forecasting the number of electoral votes a presidential candidate will receive, a bottom-up approach would be to forecast vote share in each state, and via simulation, aggregate state-level predictions for a national level prediction (Klarner 2008, 2012). Accordingly, forecasts are made for different groups of partisans here. Another reason to predict the three party registration categories separately is that some predictor variables vary for Democrats and Republicans, most importantly, the competitiveness of the two party’s primaries. Rarely, the two parties have primaries at different times in the same state and year.

Hierarchical linear modeling is used as changes in registrants for the three party registration categories may be correlated with each other in a county at one time. Many people--of whatever party--register soon before elections. Sets of error terms at the state-month and month levels are also included. The month level error term captures unmodeled factors that influence registration rates nationally.¹ Last, the rate of deadwood registrants may vary between the two parties, making adjustments for deadwood especially important when predicting election outcomes.

That new additions to the rolls goes up as elections approach is well known. How concentrated new registrations are in the months before elections and how this concentration varies from locale to locale and by party is not. Furthermore, it is not known how much new registration activity increases before primaries relative to the general election and how much the competitiveness of primaries and general elections in a state and year contribute to new adds.

¹ One complication for the hierarchical model is party switching. When some factor causes everyone to be more likely to register—such as a highly competitive general election—members of both parties will plausibly be more likely to register. But when people who are already registered switch parties, this moves the correlation between changes in the two parties in a negative direction. This may especially be the case in Iowa and Wyoming, where primary voters can choose to vote in either party’s primary, but most change the party they are registered with at the same time. If the competitiveness of the two party’s primaries is lopsided, this may cause party switching.

The variables in the model are as follows:

Total Registrants Prior Month: Total registrants by party lagged one report. This makes the model a model of change. Because the aspect of the unit of analysis representing time is so short, R-squared will be very high, but this does not imply the model is good, or bad. Prediction error two months out, three months out, etc., will compound.

Business Days in Report: Number of business days between the end date of reports.

Days in Report: Number of days between the end date of reports.

Although new registrations and drops are presumably processed on business days, the incidence of Saturdays and Sundays, holidays, etc., may give citizens time to fill out forms, etc. Saturdays, Sundays and federal holidays are not considered business days here. The number of days and business days between reports varies in small but substantial ways between reports (see Appendix).

Ban: Business Days with Ban on Purging: The NVRA prohibits removing registrants for non-voting for a period of 90 days before any primary, primary runoff, special, special primary, special primary runoff, or general election for federal offices.

Periods when purging is not permitted under the NVRA varies by county and month depending on the incidence of primaries, runoff primaries, and special elections for federal office discussed above.

Closed: Business Days Voter Registration Closed: A variable recording the number of business days between the closing date of registration and the election in question during a report, and is hypothesized to have a negative coefficient.² For election day registration states—Iowa after 2008 and Wyoming—the closing date above is considered to be the day of the election.

² When people register between an election's closing date and the election, they are still added to the rolls, they just can't vote in the upcoming election. These people are sometimes labeled "conditional voter registrations" (e.g., California). It is hypothesized that new registrations will go down during these periods, as the benefit of registering is diminished. On the other hand, people who intended to register in time for the election may do so soon after the deadline passes because they were thinking about doing it beforehand.

Election Day Registration / Election Day Registration General Election Month / Election Day Registration Primary Election Month: Bienniums in a state with election day registration are coded “1” for the entire period. Additionally, a variable coded “1” for the report that includes a general election day with EDR, as well as another variable coded “1” for a report that includes a primary election day with EDR, are also included. While EDR is known to bring up turnout generally, it may also depress registration earlier in the season, as people know they can register on election day. The EDR variables that are coded for election months only capture the effect on election day.

General Election, Presidential General Election, Presidential Primary, Primary Election, Special or Runoff Election: Dummy variables for whether the closing date for a particular type of election—general, general in a presidential year, presidential primary, primary for non-presidential offices, special elections or runoff primary elections (the latter two combined into one dummy variable)—is held during the time period covered by a report. These variables are then led between one and six months to capture the impact on registration as people anticipate voting in future elections. All of the other election variables described below that track the competitiveness of primary or general elections follow the same general coding strategy.

When a state holds its presidential primaries at the same time as primaries for other offices, both “Pri” and “PrezPri” are coded “1.” However, “PrezPri” was excluded, and is instead captured with the variable below that tracks whether a presidential nomination contest for a party was still unresolved when the presidential primary was held in a state as explained below. Why vote in a primary if your party’s nominee is already known, and the presidential primary is separate from primaries for other offices?

If special elections or runoffs only occur in a portion of the state, only the counties with these elections are coded “1.”

Monthly voter registration data are especially important to utilize because when primaries are held within a biennium (and thus registration closing dates), the institution and incidence of

primary runoffs (nine states currently, and Florida up to the 2002 election), whether states have separate primaries for president, and the incidence of special elections for federal office all vary.³

Pivotal State: A dummy variable tracking an approximate measure of whether a state is “pivotal” in the general election for president: whether vote share was within 5% of halfway between 50% of the two-party vote and the nationwide share of the two-party vote. This measure approximates how pivotal a state is--the probability a small change in the vote in a particular state will change the outcome of the election—better than a measure centered around 50% of the vote.

Competitive Gubernatorial Primary: A dummy variable tracking whether a governor’s race was won by 10% or less of the two-party vote in the general election.

Competitive U.S. Senate Primary: A dummy variable tracking whether a U.S. Senate race in the state was won by 10% or less of the two-party vote in the general election

Contested U.S. House General Election: The proportion of a county with contested U.S. House races.

Competitive U.S. House General Election: The proportion of a county with U.S. House races won by 10% or less of the vote.

Unfortunately, not much can be gleaned from the variables tracking the competitiveness of general elections because of lack of variation within states. At the extreme, Maryland, Oregon and Wyoming were never pivotal states in presidential elections, while Florida always was, and Iowa was about half the time. This means that the general election competitiveness variables are merely picking up information about conditions specific to Florida and to a lesser extent Iowa. Similar problems apply for the competitiveness of gubernatorial and U.S. Senate elections.

³ Special elections for seven U.S. House seats affecting 21 counties in different elections occurred in the time period examined here. Additionally, Alabama experienced a special election for U.S. Senate in 2017, which also had primaries and a primary runoff beforehand, creating a “no purge” window from May 17, 2017 to December 12, 2017, as well as a special election for a U.S. House seat.

Presidential Nomination Unsettled Presidential Primary: A dummy variable indicating whether a party's nomination contest for president has been settled or not by the time a state had its presidential primaries.

Competitive Gubernatorial Primary: A dummy variable indicating that a party in a state in a year had a contested primary for governor in which the winner obtained less than 55% of the vote.

Competitive U.S. Senate Primary: A dummy variable indicating that a party in a state in a year had a contested primary for U.S. Senate in which the winner obtained less than 55% of the vote.

Even though only five states are examined in this analysis, the primary competitiveness variables do exhibit a lot of variation within states over time.

For unaffiliated voter registrants, the competitiveness of the relevant office's primary for the two major parties is multiplied by "-1" and averaged together. Three of the five states analyzed are closed primary states—Florida, Maryland and Oregon. Iowa and Wyoming are generally referred to as "semi-open," because voters can choose to vote on whichever party's primary they want to on election day but must somehow publicly state they are a member of that party. However, in Iowa and Wyoming, they must take the additional step of registering with that party right then. The negative values of this variable for the unaffiliated reflect the fact that they are being "pulled into" the major parties when primaries are more competitive. If more states were in the sample, an interaction would be more appropriate to assess this effect. It should be noted that when all of the primary variables are coded "0" for the unaffiliated, the results are much the same.⁴

⁴ The fact that these variables utilize values from a future election to predict prior voter registration activity is problematic and measures based on election prediction models should be used instead.

Results

The inclusion of a lagged dependent variable means that the impact of the independent variables can be interpreted as the impact on a change in total voter registrants by party. It is important to note that all effects should essentially be multiplied by “three” to think in terms of registrants as a percentage of voting age population, as there are three party categories.

The results are reported on Table 6. The first four variables in the model capture the “mechanical” effects of business days in a report, number of days in a report, the impact of the NVRA ban on removing registrants for non-voting after failing to confirm their eligibility, and business days in a reporting period that are closed to new registrations. All of these variables had coefficients that were of the opposite sign than expected, and “business days” and “days” were statistically significant in the opposite direction.

The coefficient associated with “EDR” indicates that states with election day registration generally see less of an increase in voter registrants from month to month, but that for the report including general election day, there is a two person per 1,000 VEP higher level of increased voter registration than states with without election day registration ($p < .05$, two tailed). EDR has about half the impact for primary election months, but this variable is not statistically significant. Online voter registration is also associated with an increase in voter registrations, but this also failed to achieve statistical significance.

The impact of the general election on voter registration is clear, even during midterm elections. Except for the month including Election Day, which is often subject to registration being closed for at least a portion of the month, the three months prior to the election are associated with increases in registrants, all statistically significant at conventional levels ($p < .05$, one-tailed). Figure 1a in the Appendix shows the total effect of the general election on voter registration increase for the six months prior to the general election, showing a total impact of almost one percent by election day.

Title 6: Hierarchical Linear Model of Change in Total Voter Registrations per 1,000 Voting Age Population for Democrats, Republicans, and the Unaffiliated.

Table Variable Name	Coef.	Standard Error	P Value
Total Registrants Prior Month	0.998	0.00	0
Business Days in Report	-0.42	0.08	0
Days in Report	0.33	0.06	0
Ban: Business Days with Ban on Purging	-0.01	0.01	0.483
Closed: Business Days Voter Registration Closed	0.01	0.03	0.833
Election Day Registration	-0.42	0.22	0.054
Election Day Registration: General Election Month	2.07	0.99	0.036
Election Day Registration: Primary Election Month	0.93	0.92	0.308
Online Voter Registration	0.27	0.23	0.24
General Election: Month Of	0.50	1.13	0.656
General Election: One Month Prior	2.52	0.87	0.004
General Election: Two Months Prior	2.35	1.01	0.019
General Election: Three Months Prior	1.60	0.92	0.08
General Election: Four Months Prior	0.98	0.77	0.207
General Election: Five Months Prior	-0.30	0.89	0.738
General Election: Six Months Prior	-0.28	0.99	0.776
Presidential General Election: Month Of	-0.03	0.99	0.976
Presidential General Election: One Month Prior	4.33	0.98	0
Presidential General Election: Two Months Prior	1.56	1.20	0.194
Presidential General Election: Three Months Prior	2.39	1.17	0.04
Presidential General Election: Four Months Prior	1.81	0.97	0.062
Presidential General Election: Five Months Prior	0.96	0.99	0.333
Presidential General Election: Six Months Prior	0.53	1.21	0.661
Pivotal State: Presidential General Election: Month Of	-0.03	1.23	0.98
Pivotal State: Presidential General Election: One Month Prior	-0.23	1.22	0.848
Pivotal State: Presidential General Election: Two Months Prior	0.71	1.27	0.575
Pivotal State: Presidential General Election: Three Months Prior	-0.71	1.29	0.581
Pivotal State: Presidential General Election: Four Months Prior	-0.70	1.23	0.568
Pivotal State: Presidential General Election: Five Months Prior	0.38	1.24	0.757
Pivotal State: Presidential General Election: Six Months Prior	0.48	1.09	0.659
Competitive State: Gubernatorial General Election: Month Of	-0.90	1.23	0.466
Competitive State: Gubernatorial General Election: One Month Prior	-0.15	0.99	0.883
Competitive State: Gubernatorial General Election: Two Months Prior	-1.36	1.09	0.211
Competitive State: Gubernatorial General Election: Three Months Prior	-0.41	1.12	0.713
Competitive State: Gubernatorial General Election: Four Months Prior	0.83	1.03	0.418
Competitive State: Gubernatorial General Election: Five Months Prior	0.04	1.13	0.97
Competitive State: Gubernatorial General Election: Six Months Prior	-1.96	1.21	0.106
Competitive State: U.S. Senate General Election: Month Of	-1.86	1.41	0.187

Competitive State: U.S. Senate General Election: One Month Prior	-1.67	1.45	0.25
Competitive State: U.S. Senate General Election: Two Months Prior	0.28	1.46	0.846
Competitive State: U.S. Senate General Election: Three Months Prior	-0.63	1.45	0.665
Competitive State: U.S. Senate General Election: Four Months Prior	0.14	1.44	0.923
Competitive State: U.S. Senate General Election: Five Months Prior	0.67	1.45	0.646
Competitive State: U.S. Senate General Election: Six Months Prior	-0.69	1.39	0.621
Contested U.S. House General Election: Month Of	0.12	0.35	0.741
Contested U.S. House General Election: One Month Prior	-0.49	0.34	0.149
Contested U.S. House General Election: Two Months Prior	-0.44	0.34	0.201
Contested U.S. House General Election: Three Months Prior	-0.29	0.34	0.4
Contested U.S. House General Election: Four Months Prior	-0.60	0.34	0.079
Contested U.S. House General Election: Five Months Prior	0.15	0.34	0.659
Contested U.S. House General Election: Six Months Prior	0.10	0.34	0.77
Competitive U.S. House General Election: Month Of	-0.26	0.23	0.264
Competitive U.S. House General Election: One Month Prior	0.12	0.23	0.587
Competitive U.S. House General Election: Two Months Prior	0.11	0.23	0.639
Competitive U.S. House General Election: Three Months Prior	-0.09	0.23	0.706
Competitive U.S. House General Election: Four Months Prior	0.03	0.23	0.907
Competitive U.S. House General Election: Five Months Prior	0.15	0.23	0.508
Competitive U.S. House General Election: Six Months Prior	0.11	0.23	0.633
Primary Election: Month Of	-0.49	0.63	0.441
Primary Election: One Month Prior	0.91	0.52	0.08
Primary Election: Two Months Prior	0.82	0.50	0.1
Primary Election: Three Months Prior	0.74	0.48	0.121
Primary Election: Four Months Prior	0.55	0.38	0.145
Primary Election: Five Months Prior	0.41	0.38	0.281
Primary Election: Six Months Prior	0.16	0.31	0.613
Presidential Nomination Unsettled Presidential Primary: Month Of	0.79	0.10	0
Presidential Nomination Unsettled Presidential Primary: One Month Prior	1.21	0.09	0
Presidential Nomination Unsettled Presidential Primary: Two Months Prior	-0.01	0.09	0.877
Presidential Nomination Unsettled Presidential Primary: Three Months Prior	-0.17	0.09	0.068
Presidential Nomination Unsettled Presidential Primary: Four Months Prior	-0.47	0.09	0
Presidential Nomination Unsettled Presidential Primary: Five Months Prior	-0.36	0.09	0
Presidential Nomination Unsettled Presidential Primary: Six Months Prior	-0.26	0.09	0.006
Competitive Gubernatorial Primary: Month Of	4.79	0.13	0
Competitive Gubernatorial Primary: One Month Prior	1.30	0.13	0
Competitive Gubernatorial Primary: Two Months Prior	-0.01	0.13	0.941
Competitive Gubernatorial Primary: Three Months Prior	-0.43	0.13	0.001
Competitive Gubernatorial Primary: Four Months Prior	-0.53	0.13	0
Competitive Gubernatorial Primary: Five Months Prior	-0.35	0.13	0.008
Competitive Gubernatorial Primary: Six Months Prior	-0.30	0.13	0.022
Competitive U.S. Senate Primary: Month Of	0.84	0.13	0
Competitive U.S. Senate Primary: One Month Prior	1.49	0.13	0

Competitive U.S. Senate Primary: Two Months Prior	0.50	0.13	0
Competitive U.S. Senate Primary: Three Months Prior	-0.04	0.13	0.78
Competitive U.S. Senate Primary: Four Months Prior	1.98	0.13	0
Competitive U.S. Senate Primary: Five Months Prior	1.17	0.13	0
Competitive U.S. Senate Primary: Six Months Prior	-0.42	0.13	0.002
Special or Runoff Election: Month Of	-0.28	0.72	0.693
Special or Runoff Election: One Month Prior	0.66	0.64	0.307
Special or Runoff Election: Two Months Prior	0.95	0.63	0.133
Special or Runoff Election: Three Months Prior	0.51	0.61	0.402
Special or Runoff Election: Four Months Prior	0.70	0.59	0.235
Special or Runoff Election: Five Months Prior	-0.45	0.58	0.443
Special or Runoff Election: Six Months Prior	-0.84	0.58	0.148
Constant	-1.20	0.42	0.004
Month Error Term	.58	.17	
State-Month Error Term	2.54	.07	
County-Month Error Term	2.09	.02	
Error Term	3.76	.01	
Number of Cases = 138,996			

The additional boost to registration of a presidential election is captured by the variables labeled “Presidential General Election.” Again, the report with the general election in it is not statistically significant, but three are, which pertains to the additional boost to registration beyond a midterm election. The total impact of looming presidential elections is captured in Figure 1b (see Appendix), which shows an increase of almost 19 registrations per 1,000 VEP, more than twice that in a midterm.

None of the variables associated with the competitiveness or how pivotal a state was in presidential, gubernatorial or U.S. Senate elections is statistically significant. This may have to do with the states in this sample that happen to be competitive, which are Florida and Iowa.

The figures (see Appendix) pertaining to the impact of primary elections on change in registration by party are not labeled with specific months since states have primaries at different times. Instead, a “0” is put under the month that has the report that contains the primary election in it.

The coefficients associated with variables labeled “Primary Election” also fail to achieve statistical significance. The variables measuring the report with a presidential primary day and the month prior to that indicate a statistically significant increase in registration when there is an unsettled nomination contest for a party—Iowa comes to mind. For the first time, the variable tracking the report contemporaneous with the election shows an increase in registrants. Although the coefficients for these two variables are non-trivial, they are preceded by a number of negative coefficients, making the overall impact of a presidential primary for an “unsettled” nomination contest small (see Figure 1d).

The other sets of variables tracking competitive primaries—for Governor and U.S. Senate—show sizeable impacts, including for the report that includes primary election day. Two of the gubernatorial primary variables are statistically significant with the correct sign while five out of seven of the U.S. Senate primary variables are. One might wonder especially about the efforts that might be made to mobilize voters in a Republican U.S. Senate primary in Wyoming, where the cost to mobilize a given percentage of the electorate would be relatively low, and where an entire U.S. Senate seat is at stake—since the general election is a foregone conclusion. Last, the impact of special elections and runoff primaries was not associated with statistically significant rises in voter registration, although the five states here experienced very few of those.

Conclusion

The analysis presented in this paper has given a preliminary look at the large-scale data collection project underway, and the many applications monthly county level registration data can have. The findings about competitive primaries spurring early voter registration bring up the question of whether the upturn in voter registrations is associated with a reduction in new registrations as the election approaches—in other words, people merely registered earlier than they usually do—or whether the higher level of registration is sustained.

Another factor inhibiting the full use of these data are the ambiguities and obvious errors discussed above. However, through a combination of inquiry, fact finding and analysis, the foibles associated with the data can be purged.

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Table 5: Extreme Drops in Registrants: Florida 1997-2016 and Maryland 2002-2012

State	County	Dropped in Month as a Percentage of All Months	Year	Month	Number Dropped in Month	Number Dropped in All Years	Ave. Voter Eligible Population Across Time
FL	Lake	10.0	2005	August	12,977	129,510	199,254
FL	Volusia	10.6	2003	May	25,344	238,398	352,096
FL	Hernando	10.7	2003	May	9,120	85,609	109,943
FL	Hendry	10.8	2003	April	2,474	22,895	17,235
FL	Santa Rosa	11.2	2005	February	6,261	56,099	98,739
FL	Hamilton	11.3	2002	February	620	5,506	7,853
FL	Hendry	11.4	2001	October	2,599	22,895	17,906
FL	Putnam	11.7	2001	May	3,991	34,190	51,045
FL	Franklin	11.7	2002	February	915	7,838	6,195
FL	Seminole	11.8	2001	May	20,995	178,329	260,979
FL	Orange	12.2	2011	January	49,461	406,715	781,076
FL	Clay	12.2	2005	April	8,699	71,282	117,279
FL	Marion	12.2	2003	May	15,156	124,153	208,228
FL	Osceola	12.4	2005	August	11,044	88,816	144,879
FL	Union	12.5	2002	January	440	3,518	6,282
FL	Osceola	12.7	2001	October	11,283	88,816	119,519
FL	Columbia	12.9	2009	October	4,606	35,821	48,227
FL	Holmes	13.1	1999	February	1,107	8,472	12,529
FL	Hardee	13.3	2005	June	1,025	7,707	14,748
FL	Alachua	13.4	2003	July	14,222	105,930	169,574
FL	Jackson	13.5	2003	April	2,410	17,856	30,024
FL	Leon	14.9	2002	May	19,738	132,727	182,084
FL	Franklin	15.2	2000	February	1,191	7,838	6,542
FL	Hillsborough	16.2	2008	December	78,943	486,139	812,346
FL	Dixie	16.7	2005	August	1,189	7,123	11,004

FL	Lee	<i>17.5</i>	2016	April	45,353	259,176	520,745
FL	Miami Dade	<i>18.1</i>	2009	July	123,791	685,338	1,315,643
FL	Columbia	<i>19.5</i>	2004	November	6,979	35,821	42,573
FL	Gilchrist	<i>21.3</i>	2001	November	1,629	7,643	9,685
MD	Queen Annes	<i>10.0</i>	2009	August	1,200	12,055	34,747
MD	Carroll	<i>10.0</i>	2007	January	3,419	34,161	119,680
MD	Calvert	<i>12.3</i>	2009	February	2,356	19,197	62,844
MD	Charles	<i>12.3</i>	2009	February	4,224	34,323	101,279
MD	Worcester	<i>12.3</i>	2007	January	2,004	16,280	39,865
MD	Harford	<i>13.6</i>	2006	June	8,272	60,761	173,347
MD	Charles	<i>14.5</i>	2007	March	4,972	34,323	99,248
MD	Talbot	<i>21.1</i>	2008	March	2,607	12,383	28,732

Figure 1a: Increase in
Registrants / 1,000
VEP by Party Category Before
November Midterm Election by
Month

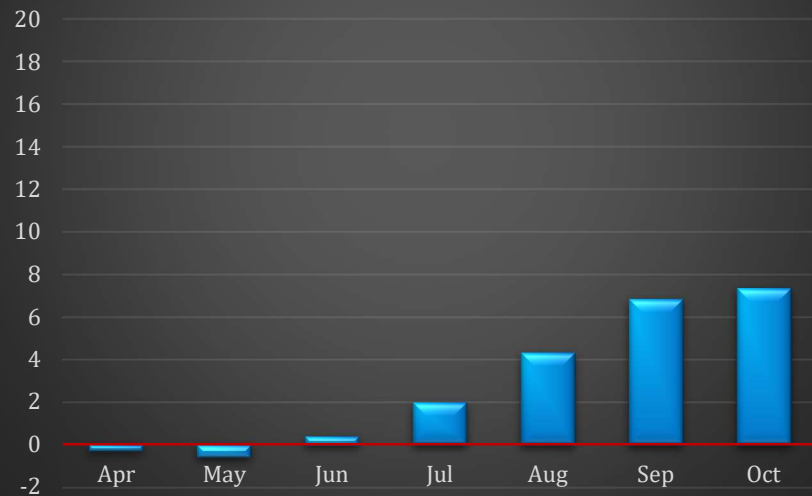


Figure 1b: Increase in
Registrants by Party Category
Before November Presidential
Election / 1,000 VEP by Month

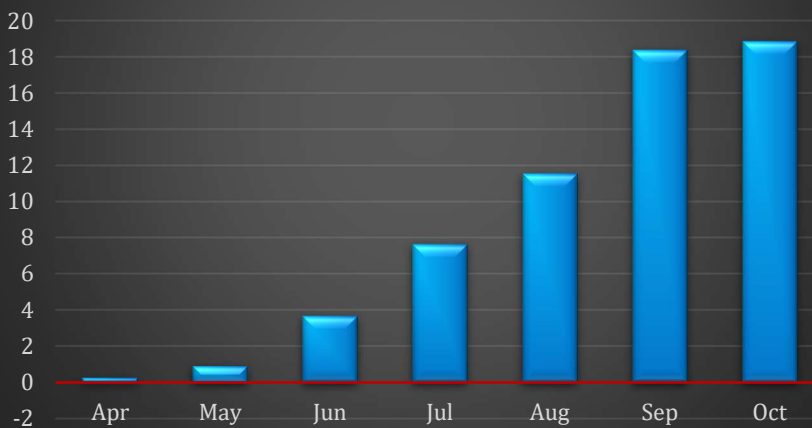


Figure 1c: Increase in
Registrants / 1,000 VEP by Party
Category Before State Primary Election
by Prior Month

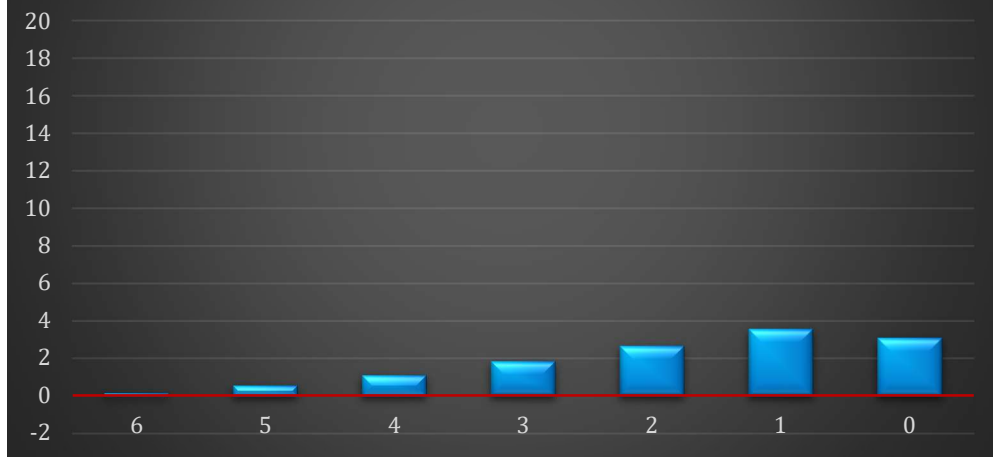


Figure 1d: Increase in
Registrants / 1,000 VEP
by Party Category Before Unsettled
Presidential Primaries Each Prior Month

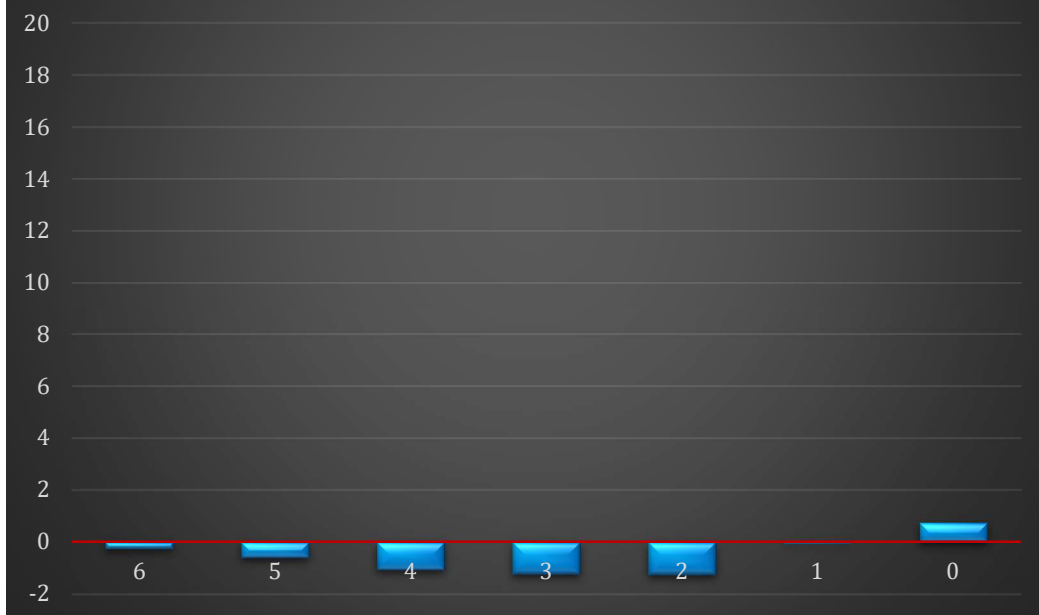


Figure 1e: Increase in Registrants / 1,000 VEP by Party Category Before Competitive Gubernatorial Primaries Each Prior Month

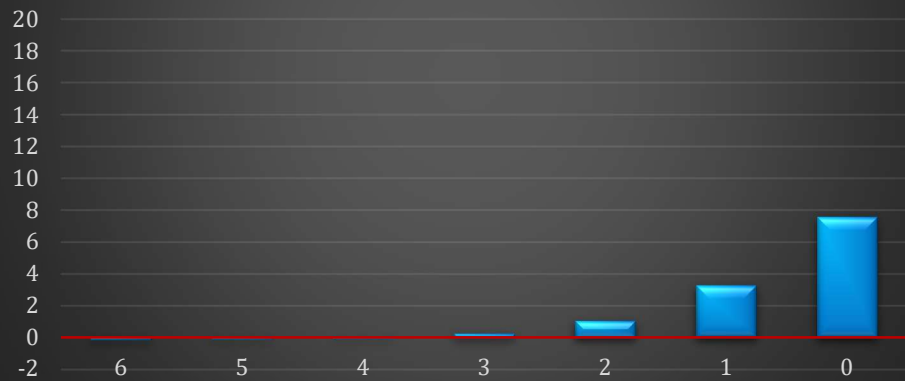


Figure 1f: Increase in Registrants / 1,000 VEP by Party Category Before Competitive U.S. Senate Primaries Each Prior Month

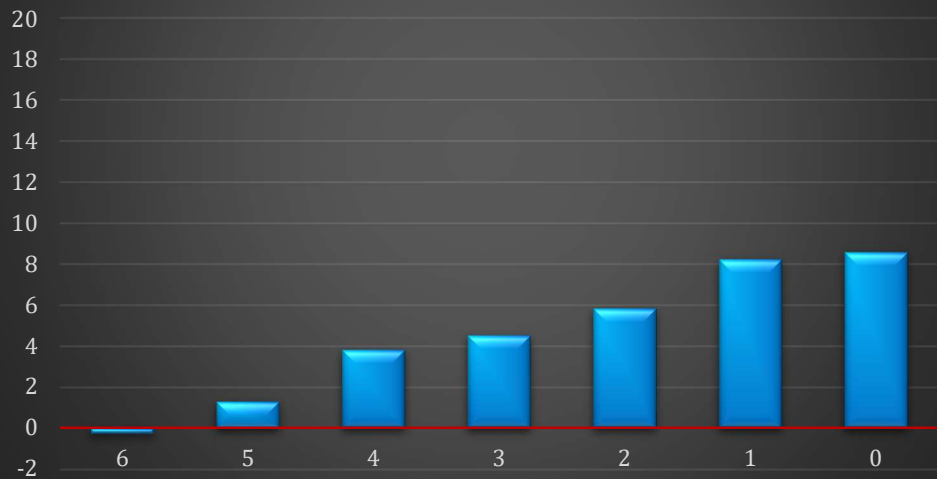
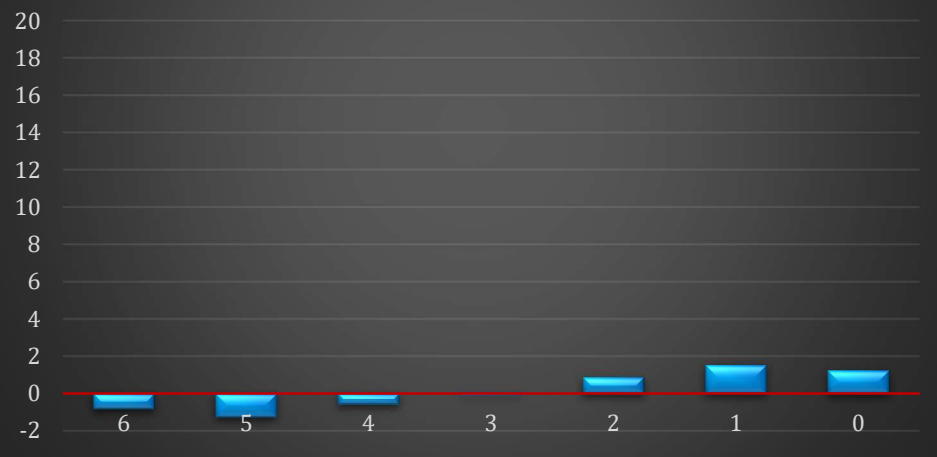


Figure 1g: Increase in
Registrants / 1,000 VEP by Party
Category Before Special or Runoff
Elections Each Prior Month



APPENDIX A

Information in Compiled Reports

State	Variables Available & Geographic Units	Years Available	Data Points
al	cy x race for status=active only, cy x status=inactive. Early reports merely have total registered.	45 months spread from 1980-1 to 2003-12	208,377
	cy x race for status=active only, cy x status=inactive	2004-11 to 2007-8	
	cy x race x active status	2007-9 to 2020-4	
fl	cy x party	1995-1 to 2016-12	319,992
	cy x drops & adds x mode	1995-1 to 2016-12	
	cy x confirmation letters / responses	1995-1 to 2010-12	
md	sld	2000, 2002, 2004, 2010, 2011, 2012 only	792,021
	council district	2012 only	
	cy x party	2000-1 to 2020-4	
	cy x party x new/dropped x mode of removal (30+ cats)	2000-1 to 2012-6	

ia	cy x active status x party	2000-1 to 2020-5	273,734
	sld x party	2008-8 to 2014-4	
	sld x active status x party	2014-5 to 2020-5	
or	cy-party	2001-1 to 2020-4	284,396
	sld x party	2001-1 to 2020-4	
	cd x party	2001-1 to 2012-10	
	cd-county x party	2012-11 to 2020-4	
wy	cy x party	2008-2 to 2020-5	20,953

Data Quality

The number of obvious errors in the data was fairly high. Examination of time series of total registrants in each county in the six states revealed frequent one-month spikes or downturns. When these were driven by a change in just one party's registrants (or just one race, in the case of Alabama), it is obvious these were typos. Elections returns on state websites are usually official historic records produced by the Secretary of State. The high error rate for voter registration figures is plausibly because monthly reports of voter registration statistics are generally not.

Up and downspikes for total registrants for one month were examined and sometimes revealed clear errors, for example, one substantial party category (Democrats, Republicans, "other party" or "no party") go up or down by several thousand for one month only, while all other party categories hardly moved at all.

Another problem is that it is unclear for some states whether their monthly reports apply to the first or last day of the month. When day of month is reported, it can tend to be towards the beginning of the month (Iowa and Wyoming) or towards the end of the month (Maryland), and is sometimes for a date late in the preceding month (Iowa). Examination of state closing dates and total registrants was used to code whether monthly reports were for the last or first day of the month when this was not reported, as inquiries to states about this matter have not been answered yet. For example, for Oregon with a registration closing date 21 days before the general election, because large increases occurred between September and October reports along with large decreases between October and November reports, it was inferred that monthly reports were for the last day of the month.

BUSINESS DAYS

The number of days voter registrations are being processed in a particular month varies in small but by surprisingly substantial amounts due to day of week on the first, the number of days and holidays in the month, the day of the month reports come out and how these all interact with election and closing dates. Further complicating this is that many states have two or more reports in one month—especially crucially important primary or general elections months. Most important, the cost of measuring business days between reports is low.